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RESEARCH ARTICLE

A Psychometric Analysis of the Duke Misophonia Questionnaire

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ABSTRACT

Background: The Duke Misophonia Questionnaire (DMQ) is one of few psychometrically constructed scales designed to measure misophonia, a condition characterized by heightened sensitivity to specific auditory stimuli, resulting in strong emotional, physiological, and behavioral responses. While prior research has evaluated the dimensionality of individual subscales, a comprehensive psychometric analysis of the entire DMQ has not been conducted.

Objectives: This study aimed to extend the psychometric evaluation of the DMQ using contemporary psychometric methodologies to enhance its consistency, stability, and applicability in future research.

Methods: Two samples of university students were recruited online via a research participation pool (Sample 1: $N = 318$; Sample 2: $N = 424$; both predominantly female and White, aged 18–30). A novel, empirically derived five-factor model (beliefs, coping, impairment, aversion, and anxiety) and the theoretically defined eight-factor model, developed from clinical and theoretical foundations, were assessed. The five-factor model was further examined using multidimensional item response theory (MIRT) to gain insights into item- and test-level functioning.

Results: Both models demonstrated adequate fit, though the five-factor model was more parsimonious and fit slightly better. Neither model contained cross-loadings. The five-factor model was further evaluated through MIRT which found that the DMQ provides the most information at moderate to high levels of misophonia severity. This analysis provided valuable information regarding the functioning and efficiency of the DMQ at both item and test levels.

Conclusion: The results contribute to the refinement of the DMQ by identifying a more parsimonious psychometric structure, promoting greater consistency and stability in its use for future misophonia research.

1 | Introduction

Misophonia is a condition characterized by a decreased tolerance for specific auditory stimuli (i.e., triggers) followed by strong, negative emotional, physiological, and behavioral responses to trigger exposure (Brout et al. 2018; Cavanna 2014; Edelstein et al. 2013; Schröder et al. 2013; Swedo et al. 2022). Triggers are primarily human-based and often, more specifically, comprised of human oral or nasal sounds (e.g., sniffing, heavy breathing, eating, chewing, or throat clearing) but also may include environmental sounds, such as those made by objects or animals (Vitoratou et al. 2021). The condition is newly described and, despite having a recent consensus definition and published sets of diagnostic criteria, formal diagnostic criteria have not been set which reduces standardization across clinical and research studies (Jager et al. 2020; Schröder et al. 2013; Swedo et al. 2022).

An additional barrier to standardization across clinical and research studies is the limited number of psychometrically developed or evaluated scales. Failing to carefully consider the psychometric quality of a measure can lead to under- or overestimation of important effects, create ambiguity or inaccuracy when interpreting scores, and lead to inaccurate conclusions about the construct(s) being measured (Borsboom 2006; Flake et al. 2022; Flake and Fried 2020; Flora and Flake 2017; Furr 2020; Horn 1965; Kyriazos and Stalikas 2018). To maximize measurement quality, researchers should only use measures with adequate psychometric evidence and construct validity. As discussed in Flake et al. (2022), there are three general phases that assist researchers in construct validation: (1) substantive, (2) structural, and (3) external. The substantive phase provides evidence of a scale's content. In the structural phase, statistical and psychometric properties of the scale are assessed. Lastly, in the external phase, researchers test whether the relationship between the instrument's scores and other constructs aligns with theory. Use of psychometrically sound instruments allows for more precise measurement, which can provide more nuanced insights into the core features of the disorder. Use of such scales will also help with efforts to identify the underlying mechanisms of misophonia.

The Duke Misophonia Questionnaire (DMQ) is an 86-item instrument developed to comprehensively assess misophonia through self-report (Rosenthal et al. 2021). DMQ development involved the refinement of a set of items drawn from other measures of misophonia (e.g., the Misophonia Questionnaire, the Amsterdam Misophonia Scale, the Misophonia Assessment Questionnaire, etc.) with input from clinicians, stakeholders, and both individuals with misophonia and their family members (Johnson 2014; Schröder et al. 2013; Wu et al. 2014). Researchers have noted the rigorous approach taken to refine the DMQ, which included both *within-subscale* exploratory factor analysis (EFA) and item response theory (IRT; Shan et al. 2024). Although the authors used these psychometric techniques while developing the DMQ, a complete dimensionality assessment has not been conducted. Rather, the researchers decided upon nine theoretically derived subscales designed to index aspects of the full misophonia latent construct (i.e., misophonia experiences/symptom categories), including (1) triggers, (2) cognitive reactions, (3) affective reactions, (4) physiological reactions, (5) beliefs about themselves and the disorder, coping responses to triggers (6) before, (7) during, and (8) after exposure, and (9) how these triggers, and subsequent reactions to them, impair quality of life.

Although Rosenthal and colleagues (2021) performed a psychometric evaluation of theoretically derived factors assessed within subscale, a dimensionality assessment was not performed on the overall scale, including all items or all subscales in a single analysis, which limits the ability of the field to determine the appropriateness of the proposed subscales despite the original fit of each subscale factor. The DMQ is now used by clinicians globally and has been translated into many languages with scoring based on the theoretical subscales (Shan et al. 2024). We argue that the factor structure of the full scale needs to be assessed before use of subscales (and scores from subscales) can be recommended. Not only would a dimensionality assessment advance the field through stronger psychometric validation of the scale, but also it could aid researchers in the development of a short-form or other efforts to expand the applications of the DMQ.

Psychometric best practices recommend the use of a combination of EFA and confirmatory factor analysis (CFA) to assess the dimensionality of a scale. Although researchers should have a priori assumptions on the *theoretical* structure of items, the dimensional structure resulting from empirical analysis of the items may differ from these assumptions. The use of factor analytic techniques aid researchers in assessing this match (or mismatch) in theoretical and empirical structures. The two main types of factor analytic techniques, EFA and CFA, are often presented in the literature as distinct techniques. However, EFA is simply a less *constrained* version of the common factor model which is commonly used in CFA applications (Jöreskog 1969). Use of EFA is preferable at early or exploratory phases of scale development to prevent both under- and overfactoring the data (Flora and Flake 2017). There may be unanticipated but substantively meaningful factors influencing subsets of items (i.e., the theoretical model underfactors), or researchers may have conceptualized an overly complicated model (i.e., the theoretical model overfactors). Psychometric best practices also argue that factor model selection should be based on the idea of simple structure, which roughly equates to each item loading strongly onto only one factor and each factor having multiple items loading strongly onto it. This allows for factor solutions to be easily interpretable, meaningful, and replicable (Thurstone 1947). Using a relatively unrestricted model, such as an EFA, may reveal that items do not follow simple structure, which can prompt for the reexamination of the number of factors (e.g., moving from two factors to three).

By using the a priori theoretically defined subscales, researchers place more constraints on the model with CFA, for example, rather than EFA. Not only are they constraining the number of factors, but by limiting analyses to within subscale, they also do not allow for interfactor correlations or cross-loadings. According to well-established psychometric best practices, researchers should not assume factors are completely independent if they are unsure how the factors relate to each other (Preacher and MacCallum 2003). Although this is not done directly and interfactor correlations were not explicitly set to zero, analyzing subscales separately fails to account for correlations between subscales which results in a loss of information (Fabrigar et al. 1999). Additionally, if using factor scores, failing to model correlations will cause the scores to become less precise. Finally, failing to model the shared variance through factor correlations can result in shared variance between factors to be absorbed down to the factor loadings. This may lead to an unnecessarily complex structure (e.g., more cross-loadings between subscales or the need for additional factors within subscale; Osborne 2015).

An EFA was not performed on the full DMQ during scale development, but rather, theory alone was used to determine the overall factor structure (Rosenthal et al. 2021). As discussed above, the initial use of a more constrained model via CFA over an EFA may have resulted in either under- or overfactoring, the latter of which is more likely. Additionally, because of fitting separate models for each subscale, factor correlations were not modeled, which has a variety of consequences, as discussed above. If failing to model relationships between subscales biased factor loadings, it is possible items were retained unnecessarily, adding length to the scale without proportional improvements in scale precision. The last concern, regarding factor loadings absorbing shared variance, seems particularly relevant for the DMQ, as Rosenthal et al. (2021) found that many of their theoretically derived subscales required the use of complex, bi-factor models.

Given these concerns, it is likely that the most parsimonious factor structure of the DMQ does not match the eight-factor, theoretically defined model. As such, the field would benefit from a complete dimensionality assessment of the DMQ. Additionally, as item analysis was only done within-factor during the development of the DMQ (Rosenthal et al. 2021), there is an opportunity for a more detailed item analysis that incorporates the multidimensional structure of the scale, such as MIRT.

According to Edwards (2009), IRT is a collection of latent variable models that seek to uncover the underlying process that influences responses to observed variables. This relationship is driven by properties of individuals (i.e., θ, which corresponds to level of the latent variable or construct of interest such as misophonia in the present context) and items. IRT is a preferred method because it extracts more detailed parameters about items than other psychometric techniques, such as factor analysis. Additionally, scoring is an inherent part of IRT models, and IRT scale scores offer a number of benefits including a common, easily interpretable metric, greater score variability than summed scores, conditional standard errors, and straightforward equating between samples (de Ayala 2009; Embretson and Reise 2000). Although traditional IRT models common in the psychological sciences (e.g., Samejima's graded response model [GRM]; 1997) assume unidimensionality, researchers have developed multidimensional adaptations of these models (e.g., multidimensional graded response model [MGRM]; see De Ayala 1994) that are able to account for correlations between factors.

Additional benefits of working under the IRT framework for scale evaluation can be found in Edwards (2009), Embretson and Reise (2000), and Thissen and Steinberg (2009). For the purposes of this study, we were most interested in the benefits associated with scoring and score precision. Scoring in IRT involves weighting response patterns using the item parameters instead of weighting all items equally, as is done for summed scores. IRT scale scores are also more variable and allow for a greater degree of differentiation between individuals. Score precision, closely related to reliability, is also approached differently in IRT than through a typical summed score approach. Rather than assuming all scores are equally reliable, an untenable assumption in almost all cases, precision/reliability is variable in IRT. IRT characterizes precision/reliability using several different metrics, including Fisher information (hereafter called information). Information is provided at the item level through item information functions (IIFs) as well as at the test level through test information functions (TIFs). Because of the benefits of IRT, an item-level analysis using MIRT may provide the field with new information about the DMQ items that could help the field with further refining the scale or developing a short form.

The purpose of the current study was to evaluate the dimensionality of the DMQ using methods guided by the psychometric literature. Specifically, the primary aims were to (1) psychometrically assess the dimensionality of the DMQ to determine its factor structure using an EFA/CFA approach, (2) given the empirically derived structure of the DMQ is multidimensional, perform an exploratory item analysis via the MIRT framework, and (3) perform the first evaluation of the theory-based, eight-factor structure of the DMQ via psychometric techniques (CFA). Additionally, we compared the dimensionality of the theory-defined, original eight-factor model against the most plausible model from our approach guided by the psychometric literature. Finally, we performed content analysis to better understand the resulting factors and used regression analysis to provide evidence for validity.

2 | Methods

2.1 | Participants

The data for the current study were collected via the University of Oklahoma's Psychology Research Pool (SONA). The SONA system is a secure online participation system through the university's psychology department. All data were collected anonymously via Qualtrics and those who completed the survey were compensated with 0.5 SONA credits or class credit. Given data collection occurred online, two scale-specific attention checks were used to ensure that the data obtained were valid. All study procedures were approved by the Institutional Review Board (IRB). All participants electronically acknowledged their informed consent to participate before completing the survey. The SONA system procedures require researchers to split recruitment across two time windows. Thus, for the purposes of the current study, the time windows were used to split the sample into an exploratory set (the first time window; Sample 1) and validation set (the second time window; Sample 2).

No missing data were present in either Sample 1 or Sample 2. After removing participants who (1) failed at least one of the attention checks, or (2) were not at least 18 years of age, Sample 1 contained 318 participants. Men were found to be less likely to pass attention checks than women ($\beta = -0.94$, $SE = 0.28$, $z = -3.34$, $p < 0.001$). Participants ($N = 318$) were predominantly female (78.9%; consistent with other studies where participants opt into participation [Naylor et al. 2021; Rosenthal and Rosnow 2009; Wu et al. 2014; Yektatalab et al. 2022; Zhou et al. 2017]), White (60.7%), and ranged from 18 to 30 years of age ($M = 18.7$, SD $= 1.40$).

Similar to Sample 1, we removed participants from Sample 2 who (1) failed at least one of the attention checks or (2) were not at least 18 years of age, or (3) had previously participated in data collection. A final validation sample size of 424 was obtained. Men and women were not statistically different in their likelihood of passing attention checks ($\beta = 0.38$, $SE = 0.35$, $z = 1.098$, $p > 0.05$). Participants ($N = 424$) were predominantly female (77.8%; consistent with other studies where participants opt into participation), White (59.7%), and ranged from 18 to 26 years of age ($M = 18.9$, SD = 1.18).

2.2 | Measures

2.2.1 | DMQ

The full DMQ is an 86-item, self-administered measure that consists of nine theoretically derived (rather than empirically derived) dimensions; affective responses (eight items), physical responses (five items), cognitive responses (10 items), coping-before the event (six items), coping-during the event (10 items), coping-after the event (five items), impairment (12 items), beliefs (14 items), and triggers (17 items). One trigger item asking about how bothered participants are on average by their triggers is answered on a 6-point Likert-type scale while all other trigger items are binary. The remaining items are answered with a 5-point Likert-type scale, to assess the frequency, such as: (0) never, (1) rarely, (2) sometimes, (3) often, (4) always/almost always. Rosenthal et al. (2021) did not assess the overall factor structure of the scale, but rather, assessed model fit within subscale and found all subscales fit well. Although the original 86-item scale contains trigger items, these items are (a) measured on a binary yes/no scale and (b) not meant to assess the misophonia construct, but rather, simply get participants to think about these triggers before answering construct related questions. As such, these questions were left out of the subsequent psychometric analysis (i.e., only the nontrigger items were included).

2.2.2 | Validation Measures

Three measures were used to provide preliminary validity evidence for the most plausible DMQ factor structure found. The measures are described in detail below.

Misophonia Questionnaire: The misophonia questionnaire (MQ) is a three-part, 20-item self-report measure designed to index misophonia symptoms (Wu et al. 2014). The first part (eight items) assesses specific auditory trigger categories known to be associated with misophonia. The second part (11 items) evaluates the emotions and behaviors associated with trigger exposure. The third part is a single item measure which asks participants to rate the severity of their sensitivity to auditory triggers, considering the number of sounds, level of distress, and life impairment resulting from sound sensitivities on a scale from 1 (“minimal”) to 15 (“very severe”). The symptoms scale demonstrated good internal consistency ($\alpha = 0.86$) in Wu et al. (2014), Sample 1 ($\alpha = 0.79$ and $\omega = 0.79$), and Sample 2 ($\alpha = 0.80$ and $\omega = 0.81$). The emotions and behaviors scale also demonstrated good consistency ($\alpha = 0.86$) in Wu et al. (2014), Sample 1 ($\alpha = 0.83$ and $\omega = 0.83$), and Sample 2 ($\alpha = 0.88$ and $\omega = 0.88$).

The MQ severity question is traditionally used to indicate the presence of misophonia, splitting participants into groups reflecting clinically significant levels of misophonia based on a threshold cutoff value of seven and above (Wu et al. 2014). The MQ is one of the most commonly used scales to assess misophonia, making the MQ a good option for assessing convergent validity (Brout et al. 2018; Potgieter et al. 2019).

Adolescent and Adult Sensory Profile: The adolescent and adult sensory profile (ASP) is a self-report six part, four factor measure of sensory processing patterns and effects on function performance (Brown et al. 2001). The ASP indexes individual responses to sensations, as opposed to an individual's general response or cognitive appraisal of a stimulus. The four factors are low registration (15 items), sensory sensitivity (15 items), sensation avoiding (15 items), and sensation seeking (15 items). All factors demonstrated good internal consistency ($\alpha > 0.75$) except the sensation seeking factor ($\alpha > 0.60$) in Brown et al. (2001). In Sample 1 ($\alpha > 0.70$ and $\omega > 0.74$) and Sample 2 ($\alpha > 0.73$ and $\omega > 0.77$), all factors demonstrated good internal consistency.

Anxiety Symptoms Questionnaire: The anxiety symptoms questionnaire (ASQ) is a self-report measure of the frequency and severity of anxiety-related symptoms (Baker et al. 2019). Participants are asked to rate the extent to which they have experienced various anxiety symptoms over the past week on a Likert-type scale ranging from 0 (Never) to 10 (Very Much). The ASQ includes items that capture both cognitive (e.g., excessive worry, difficulty concentrating) and somatic (e.g., muscle tension, restlessness) aspects of anxiety. Higher scores indicate greater anxiety symptoms. The ASQ has demonstrated strong internal consistency ($\alpha = 0.96$) in Baker et al. (2019) and has been validated against other established measures of anxiety, supporting its use in both clinical and nonclinical populations. The scale also demonstrated good internal consistency in Sample 1 ($\alpha = 0.96$ and $\omega = 0.97$) and Sample 2 ($\alpha = 0.97$ and $\omega = 0.97$).

2.3 | Psychometric Analysis

2.3.1 | Dimensionality Assessment

The current study assessed the dimensionality of the DMQ using a combination of EFA and CFA. Sample 1 was used in the exploratory phase to find plausible models through both EFA and CFA. In the confirmatory phase, only CFAs were fit to Sample 2, a sample that was held out of any preceding analyses to validate the model(s) identified in the exploratory phase. Given that the model(s) from the exploratory phase were validated in the confirmatory phase, Sample 1 and Sample 2 were then be combined to obtain final estimates.

Exploratory Factor Analysis: As mechanical rules, such as the eigenvalue greater than one rule (K1) or parallel analysis, tend to be arbitrary and produce poor results alone, multiple methods for determining the number of factors were applied (Fabrigar et al. 1999). Based on recommendations from Fabrigar et al. (1999), the number of factors were determined by both parallel analysis and the scree plot which have been shown to outperform other methods (Horn 1965; Manapat et al. 2025). All EFAs were conducted using the *psych* package (Revell 2024) in

R (R Core Team 2024). The exploratory Sample 1 was used for all EFAs. Since the prior study evaluating the DMQ did not run an EFA, this study is the first to conduct an exploratory (i.e., blind) dimensionality assessment of the scale (Rosenthal et al. 2021). We accounted for the categorical nature of the items using polychoric correlations and ordinary least squares estimation (Edwards 2009). The polychoric correlations between the 70 items for Sample 1 can be found in the Supporting Information. For all models, oblique rotation was used given that orthogonal rotations, which do not allow for correlations between factors, are rarely justifiable in psychological science (Fabrigar et al. 1999). Direct quartimin was selected for rotation as it has been seen to successfully produce interpretable solutions (Fabrigar et al. 1999; Manapat et al. 2025).

Confirmatory Factor Analysis: Factor structures deemed plausible using EFA in Sample 1 were first fit as a CFA in Sample 1 (exploratory CFAs) using the *lavaan* package (Rosseel 2012) in R (R Core Team 2024). Then, this smaller subset of plausible models from Sample 1 were fit using CFAs in Sample 2 (confirmatory CFAs). Final estimates were obtained by fitting a CFA to the combined Sample 1 and Sample 2 ($N = 742$; hereafter referred to as the full sample). For all CFAs, factor variances were set to one for model identification. Polychoric correlations and the diagonally weighted least squares (DWLS; also called WLSMV in *lavaan*) estimator were specified to account for the ordinal nature of the Likert-type items and to produce accurate model fit indices (Christoffersson 1975; Flora 2020). As was the case with EFAs, no previous full-scale CFA has been fit to the DMQ, with Rosenthal et al. (2021) having separated the scale into subscales according to a priori theoretical item content groupings and then analyzing each subscale separately using IRT and factor analysis. As described above, CFA models fit using Sample 1 were validated using Sample 2. Standard model fit indices including the χ^2, Tucker–Lewis index (TLI), comparative fit index (CFI), standardized root mean square residual (SRMR), and root mean square error of approximation (RMSEA) were used to evaluate all CFAs. Hu and Bentler (1999) suggest that acceptable model fit includes TLI and CFI values greater than 0.90 and an SRMR less than 0.08. Browne and Cudeck (1992) suggest an acceptable RMSEA is less than 0.10. Given that χ^2 statistics are dependent on sample size, it is possible that a representative model be rejected due to a large sample size alone. As such, researchers suggest making decisions about model fit using a combination of indices (Goretzko et al. 2024).

2.3.2 | Item Analysis

When EFA and CFA suggest multidimensionality, there are two suggested IRT approaches (Edwards 2009). The first approach suggests modifications to the scale (e.g., dropping items) to achieve a plausible unidimensional scale. The second approach is to model the multidimensionality using MIRT. Constructs are typically multidimensional in psychology, and MIRT methods allow for proper modeling of such scales, again, by incorporating the relationships between factors (Ackerman et al. 2003; Monroe and Cai 2015; Reckase 1997; Wirth and Edwards 2007). If the dimensionality assessment supports a multidimensional structure, MIRT will be used over unidimensional IRT. As such, an MGRM would be fit to the full sample, rather than a GRM, based on the model deemed most plausible by EFA and CFA.

Multidimensional Graded Response Model: Samejima's (1997) GRM is the appropriate IRT model for psychological scales with items that are measured using an ordered scale, such as a Likert-type scale. The unidimensional GRM takes the form:

$$P(x_{ij} = c|\theta) = \frac{1}{1 + \exp[-a_j(\theta_i - b_{jc})]} - \frac{1}{1 + \exp[-a_j(\theta_i - b_{j(c+1)})]}$$

which represents the probability of endorsing response option c given the latent trait (θ) and produces the observed response (x_j) to item j. For more on the GRM and IRT, in general, see Edwards (2009) and Embretson and Reise (2000).

The GRM has been extended to incorporate multiple dimensions of latent traits, resulting in the MGRM. In this extension, the probability of endorsing a response option (c) is influenced by a vector of latent traits, $\theta_i = (\theta_1, \theta_2, \ldots, \theta_k)$ and the discrimination parameters, a_j, become vectors reflecting the relationship between each dimension and the item. The item parameters for the MGRM were estimated using the *mirt* package (Chalmers 2012) in R (R Core Team 2024). The analyses under the MGRM were conducted on the full sample. Estimation was performed using the Metropolis-Hastings Robbins-Monro (MH-RM) algorithm (Cai 2010a, 2010b). All other program specifications were set to defaults.

2.3.3 | Internal Consistency

Coefficient alpha (Cronbach 1951) and coefficient omega (McDonald 1970) were computed using the *coefficientalpha* package (Zhang and Yuan 2023) in R (R Core Team 2024). It is common practice to report coefficient alpha, which is why we have included it here. However, coefficient alpha (unless strict, almost unrealistic test assumptions like tau-equivalence are met) is a lower bound of reliability (Sijtsma 2008). Therefore, a more reasonable estimate of reliability is coefficient omega which provides good reliability estimates with more realistic test assumptions (McNeish 2018).

2.3.4 | Validation Evidence

IRT scale scores were calculated using the full sample. Specifically, expected *a posteriori* (EAP) estimates, derived from the mean of the posterior distribution, were obtained for each factor using the *mirt* packgage in R (Chalmers 2012; Thissen and Wainer 2001). The EAPs for each factor of the most plausible model were entered simultaneously as predictors of the MQ subscales to assess convergent validity. The EAPs for each factor of the final model were entered simultaneously as predictors of both the ASQ and ASP (separate models for each outcome) to assess divergent validity.

3 | Results

Determining the factor structure of the DMQ was first guided by a set of EFAs conducted on Sample 1. The number of factors to estimate was determined using a scree plot and parallel analysis (Supporting Information S1: Figure S1). The scree plot suggested two or three factors and the parallel analysis suggested eight factors. As a result, we estimated models with two through eight factors.

Solution selection among these models was based on (a) the idea of simple structure (Thurstone 1947), (b) retention of as many items as possible, thus maintaining maximal information, and (c) construct validity. A solution with simple structure has high variability in loadings within factors and low factorial complexity (i.e., it minimizes cross-loadings between factors), which allows factor solutions to be easily interpretable, meaningful, and replicable. The two-, three-, four-, six-, seven-, and eight-factor empirically derived models (a) failed to maintain simple structure and (b) resulted in the loss of many items (when requiring a loading of at least 0.32). Specifically, the two-factor model resulted in the loss of two items and contained five items with cross-loadings. The three-factor model retained all items but contained 13 items with cross-loadings. The four-factor model retained all items but contained 22 items with cross-loadings. The six-factor model resulted in the loss of one item and contained 11 items with cross-loadings. The seven-factor model resulted in the loss of two items and contained five items with cross-loadings. The eight-factor model, which did not match the theoretical model conceptualized in Rosenthal et al. (2021), resulted in the loss of three items and contained five items with cross-loadings. The five-factor model, on the other hand, contained no cross-loadings and retained all items.

The two-, three-, four-, six-, seven-, and eight-factor models were deemed implausible as they resulted in item loss and contained cross-loadings. Thus, we focused our attention on the five-factor model. Factor loadings, communalities, and factor correlations for the five-factor model can be found in Supporting Information S1: Table S1. The 5F model was then examined via CFA in Sample 1, and as expected, the model fit well, so we proceeded to fit the five-factor model in Sample 2 for validation.

The 5F model adequately fit the data $\chi^2(2335) = 4916.518$, $p < 0.0001$; TLI = 0.942; CFI = 0.944; SRMR = 0.073; RMSEA = 0.051, 90% confidence interval (CI) [0.049, 0.053]. Because this factor structure held in Sample 2, Sample 1 and Sample 2 were combined to obtain final estimates for the 5F model. The factor loadings and factor correlations for the final 5F model are reported in Table 1.

Content analysis was then performed on the 5F model to better understand the clinical implications of the five factors. The first factor mapped directly onto the beliefs factor from the original DMQ paper, and as such, we retained the original label for the items associated with this factor: "beliefs" (Rosenthal et al. 2021). The second factor was comprised of all coping related items except one "during" coping item and thus we labeled this factor "coping." The third factor contained all impairment items and one item from the cognitive scale that read, "I thought about physically hurting the person making the sound." As causing another individual physical pain likely has bearing on quality of life, we named this factor "impairment." Additionally, thoughts about hurting others do not necessarily map directly onto physically harming others which means the item may index a distressing thought pattern that interferes with daily living rather than an externalizing symptom. The fourth factor contained five affective scale items that centered on negative emotional experiences, seven cognitive scale items that index negative thought patterns and thoughts on avoiding or stopping the sound, and one coping during scale item that reads, "I blocked the sound (e.g., covered ears with hands, headphones, ear plugs)." The items in factor four largely represent negative emotionality and aversion, including behavioral strategies for either avoiding or stopping the sound; thus, we labeled it "aversion." The fifth factor contained three affective scale items, five physiological scale items, and two cognitive scale items. All items in factor five were related to facets of anxiety, where the affective factor contained items gauging worry, anxiousness, and feelings of "jitteriness." The physiological scale items indexed rigidity and somatic experiences associated with autonomic arousal. Finally, the cognitive scale items were "I am helpless" and "I want to cry" which may largely represent emotional experiences associated with feelings of fear and anxiety. Given the items in factor five center on negative affect, and more specifically on feelings of anxiety and powerlessness, we labeled it "anxiety."

Given the evidence presented and that psychometric best practices support the 5F model, a five-factor MGRM was performed using the full sample. The slope (a) and threshold (b) parameters for the five-factor model are provided in Table 2. Variability in the slopes indicates that the items of the DMQ were differentially related to the misophonia construct. Using Items 1 and 3 as examples, the slope is higher for Item 1 than for Item 3 which can be interpreted to mean that Item 1 provides more information about an individual's level of misophonia than Item 3 (depicted in Figure 1 through a higher peak in the trace lines for Item 1 than the trace lines for Item 3). Variability can also be easily observed in the thresholds. For example, the thresholds for responding 'Rarely' to Item 2 was less than the thresholds for the same response on Item 12 indicating that an individual requires a higher level of misophonia to respond 'Rarely' to Item 12 than Item 2. This is illustrated in Figure 2, as the trace lines for Item 12 are shifted to the right of the θ continuum.

Information does not have a straightforward interpretation in multidimensional models; however, information can be approximated with five, separate unidimensional GRMs. Because each item loads onto only one factor, indicating between-item multidimensionality, the set of items for the beliefs factor could be modeled in one unidimensional IRT model and the set of items for the coping factor could be modeled in another, and so on. Between the multidimensional solutions and the unidimensional approximations, the EAP scores were correlated 0.97 for beliefs, 0.97 for coping, 0.98 for impairment, 0.97 for aversion, and 0.98 for anxiety. Therefore, it is reasonable to use information from the unidimensional models for interpretation. The approximated information functions are plotted by factor in Figure 3. The thresholds determine where the peak of the information function is located

TABLE 1 | Confirmatory factor analysis: Factor loadings and factor correlations for the five-factor model of the 70-item DMQ (full sample; $N = 742$).

	Factor				
Item	**Beliefs**	**Coping**	**Impairment**	**Aversion**	**Anxiety**
57	1.00				
58	1.00				
59	1.06				
60	0.97				
61	0.94				
62	0.84				
63	0.95				
64	1.03				
65	0.96				
66	0.97				
67	1.03				
68	1.04				
69	1.05				
70	1.03				
24		1.00			
25		0.96			
26		1.09			
27		1.16			
28		1.05			
29		1.11			
31		1.12			
32		1.01			
33		0.95			
34		0.72			
35		1.01			
36		0.88			
37		0.85			
38		0.99			
39		0.83			
40		1.03			
41		0.96			
42		1.08			
43		1.04			
44		1.09			
23			1.00		
45			1.14		
46			1.01		
47			1.11		
48			1.15		
49			1.11		
50			1.17		

(Continues)

TABLE 1 | (Continued)

Item	Factor				
	Beliefs	Coping	Impairment	Aversion	Anxiety
51			1.20		
52			1.16		
53			1.16		
54			1.17		
55			1.02		
56			1.18		
1				1.00	
3				0.74	
4				1.12	
6				1.14	
8				1.04	
16				1.01	
17				1.35	
18				1.22	
19				1.25	
20				1.27	
21				1.14	
22				1.10	
30				1.07	
2					1.00
5					1.17
7					0.95
9					1.04
10					0.90
11					1.17
12					1.19
13					0.98
14					1.21
15					1.20

Factor	Factor correlations				
	1	2	3	4	5
Beliefs	1				
Coping	0.42	1			
Impairment	0.52	0.36	1		
Aversion	0.39	0.35	0.32	1	
Anxiety	0.46	0.34	0.35	0.34	1

Note: Full wording of each item can be found in Supporting Information S1: Appendix A.

(e.g., if the thresholds are higher, the information function will peak further right indicating that the DMQ is more precise for respondents with higher levels of misophonia). If the slopes are higher, there is more area under the TIF, meaning that the DMQ will be more precise, on average, across the full range of misophonia.

Examination of the information functions for the DMQ indicates that this scale is more precise at higher levels of misophonia, meaning that it is more capable of distinguishing between moderate to severe levels of misophonia than low to moderate. However, this varies within factor. For example, the aversion factor is evenly distributed across the range of

TABLE 2 | Multidimensional graded response model parameter estimates for the five-factor model of the 70-item DMQ (full sample; $N = 742$).

Item	a_1	a_2	a_3	a_4	a_5	b_1	b_2	b_3	b_4
57	2.779					0.503	1.028	1.637	2.287
58	2.918					0.597	1.160	1.686	2.443
59	4.577					0.976	1.441	1.913	2.604
60	2.719					0.561	1.119	1.632	2.218
61	2.598					0.401	0.949	1.499	2.108
62	1.886					−0.162	0.442	1.319	2.425
63	2.443					0.659	1.203	1.877	2.705
64	3.910					0.975	1.475	1.888	2.461
65	2.607					0.320	0.836	1.452	2.230
66	2.996					0.982	1.439	1.910	2.433
67	3.926					0.796	1.241	1.708	2.302
68	4.000					0.878	1.294	1.781	2.327
69	4.527					0.949	1.392	1.799	2.437
70	3.510					0.980	1.421	1.890	2.546
24		1.662				−0.324	0.546	1.472	2.455
25		1.666				−0.970	−0.280	0.697	1.821
26		2.764				−0.242	0.415	1.053	1.878
27		2.290				−0.062	0.709	1.500	2.293
28		2.232				−0.601	−0.040	0.855	1.958
29		2.577				0.022	0.639	1.359	2.152
31		2.639				−0.232	0.421	1.198	2.108
32		1.971				−1.207	−0.528	0.454	1.588
33		1.637				0.036	0.802	1.571	2.710
34		1.092				−0.741	0.264	1.336	2.647
35		1.815				−1.042	−0.311	0.661	1.609
36		1.723				−0.134	0.740	1.812	2.934
37		1.389				−0.936	−0.060	0.898	2.425
38		1.679				−1.467	−0.676	0.284	1.427
39		1.617				−0.201	0.608	1.508	2.648
40		1.995				−0.318	0.453	1.255	2.249
41		1.735				−0.591	0.058	0.880	1.791
42		2.459				0.074	0.696	1.417	2.300
43		2.060				0.203	0.812	1.477	2.408
44		2.278				0.064	0.702	1.390	2.246
23			1.314			1.177	1.930	2.612	3.250
45			2.659			0.259	1.025	1.841	2.679
46			1.727			−0.077	0.847	1.757	2.886
47			2.728			0.751	1.350	1.997	2.806
48			2.573			0.633	1.364	2.098	3.163
49			2.674			0.448	1.098	1.693	2.364
50			3.147			0.121	0.828	1.530	2.440
51			3.357			0.986	1.408	1.953	2.377
52			3.769			1.177	1.469	1.953	2.440

(Continues)

TABLE 2 | (Continued)

Item	a_1	a_2	a_3	a_4	a_5	b_1	b_2	b_3	b_4
53			3.719			0.668	1.257	1.874	2.384
54			3.800			0.563	1.208	1.802	2.375
55			2.151			0.007	0.725	1.475	2.160
56			3.127			0.713	1.329	1.779	2.261
1				1.903		−1.384	−0.350	0.646	1.923
3				1.025		−1.642	−0.479	0.913	2.669
4				1.863		−0.334	0.620	1.393	2.411
6				1.762		−0.110	0.699	1.548	2.588
8				1.874		−1.655	−0.984	0.085	1.203
16				1.719		−1.224	−0.624	0.199	1.399
17				2.245		0.225	0.876	1.433	2.186
18				2.316		−0.649	−0.006	0.619	1.616
19				2.640		−1.227	−0.558	0.249	1.136
20				2.669		−0.668	0.067	0.697	1.448
21				2.226		−0.433	0.148	0.836	1.484
22				1.473		0.249	1.022	1.692	2.480
30				1.192		−1.385	−0.559	0.469	1.875
2					1.875	−0.779	0.095	0.930	2.095
5					2.711	0.142	0.852	1.409	2.099
7					1.782	−0.376	0.299	1.148	2.158
9					1.593	−0.255	0.635	1.406	2.707
10					1.590	0.051	0.887	1.694	2.649
11					3.077	0.101	0.832	1.454	2.241
12					2.636	0.239	0.897	1.526	2.433
13					1.763	0.181	0.974	1.854	2.986
14					2.045	0.787	1.470	2.076	2.976
15					2.162	0.156	0.725	1.325	2.290

Note: Full wording of each item can be found in Supporting Information S1: Appendix A.

misophonia while the beliefs factor provides little to no information at lower levels of misophonia (Figure 3). In general, the DMQ functions less well for those with low levels of misophonia. Rosenthal et al.'s (2021) goal in development of the DMQ was to develop a psychometrically valid self-report measure of misophonia to address the “lack of rigorously validated assessment measures” (p. 1). As such, this range of precision seems appropriate. However, if individuals were interested in expanding the scale so it measures both the construct of misophonia as well as threshold clinical misophonia, they may want to target item development to items that provide more information for lower levels of misophonia.

Measures of internal consistency were computed for the DMQ using the 5F model on the full sample. Coefficients alpha (Cronbach 1951) and omega (McDonald 1970) for the beliefs factor were 0.94 and 0.93, respectively. For the coping factor, alpha and omega were both 0.95. For the impairment factor, alpha and omega were both 0.93. For aversion, alpha was 0.91 and omega was 0.92. For anxiety, alpha and omega were both 0.91.

3.1 | Validation Evidence for the DMQ

Convergent and divergent validity were assessed via regression analysis (Table 3). Regarding convergent validity, we were most interested in the relationship between the MQ and the DMQ because the MQ is the most commonly used measure of misophonia. In this model, we found that all DMQ factors except anxiety were significant predictors of the MQ severity score, all DMQ factors were significant predictors of the MQ emotions and behaviors subscale score, and all DMQ factors except beliefs were significant predictors of the MQ symptoms subscale score. While most factors of the 5F model exhibit convergent validity, some factors do not. As such, the 5F model of the DMQ differentially relates to various MQ subscales.

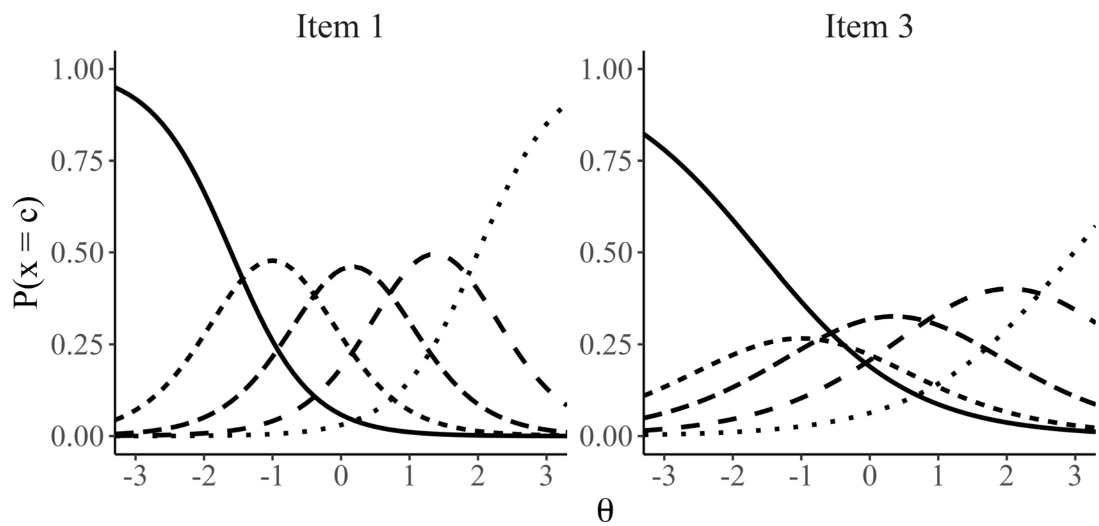

FIGURE 1 | Trace lines for Items 1 and 3 associated with the aversion factor, modeled using the 5-factor MGRM of the DMQ. Each trace line represents the probability P(x = c) of a response in category c, plotted against the latent trait θ. The solid, dashed, and dotted lines correspond to different response categories, illustrating how the probability of endorsing each category (or higher) varies across levels of the latent trait. The panels for Item 1 and Item 3 allow for comparison of their response category probabilities as functions of θ, providing insight into the distinct patterns of discrimination and category thresholds for these items. Item 1 has a larger slope (the curves are more peaked) indicating that it provides more information about misophonia than Item 3. DMQ, Duke Misophonia Questionnaire; MGRM, multidimensional graded response model.

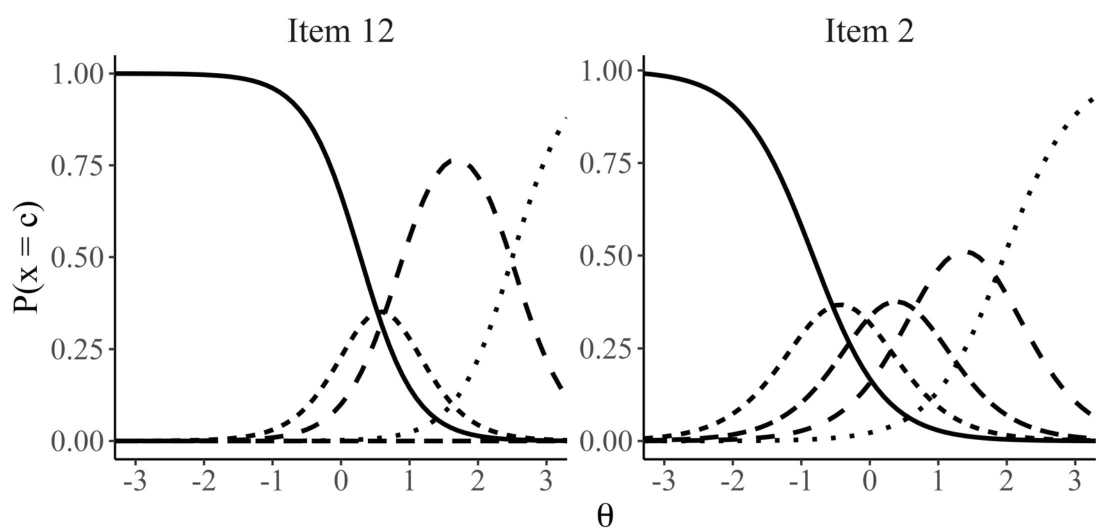

FIGURE 2 | Trace lines for Items 2 and 12 associated with the anxiety factor, modeled using the 5-factor MGRM of the DMQ. Each trace line represents the probability P(x = c) of a response in category c, plotted against the latent trait θ. The solid, dashed, and dotted lines correspond to different response categories, illustrating how the probability of endorsing each category (or higher) varies across levels of the latent trait. The panels for Item 2 and Item 12 allow for comparison of their response category probabilities as functions of θ, providing insight into the distinct patterns of discrimination and category thresholds for these items. Item 12 has a higher threshold for the "Rarely" response option than Item 2, indicating that individuals need to have a greater amount of misophonia to respond "Rarely" to Item 12 than to Item 2.

Regarding divergent validity, we were interested in the DMQ as it relates to both the ASQ and ASP. When evaluating divergent validity against anxiety (measured by ASQ), all factors except the coping factor of the DMQ were found to be significant predictors of the ASQ total score. We used the significant relationship between the ASQ and the "anxiety" factor as further justification for labeling the factor "anxiety" over "negative affect." Given this significant relationship, however, we estimated semipartial correlations to get a sense for how much variance may be left unexplained. We found that 77% of the variance in DMQ anxiety is not accounted for by generalized anxiety.

When evaluating divergent validity against sensory processing disruptions (measured by ASP), we found the impairment, aversion, and anxiety factors were significant predictors of the low registration factor. The coping factor was a significant predictor of the sensation seeking factor. Beliefs, impairment, aversion, and anxiety factors were all significant predictors of the sensory sensitivity factor. Finally, the beliefs, coping,

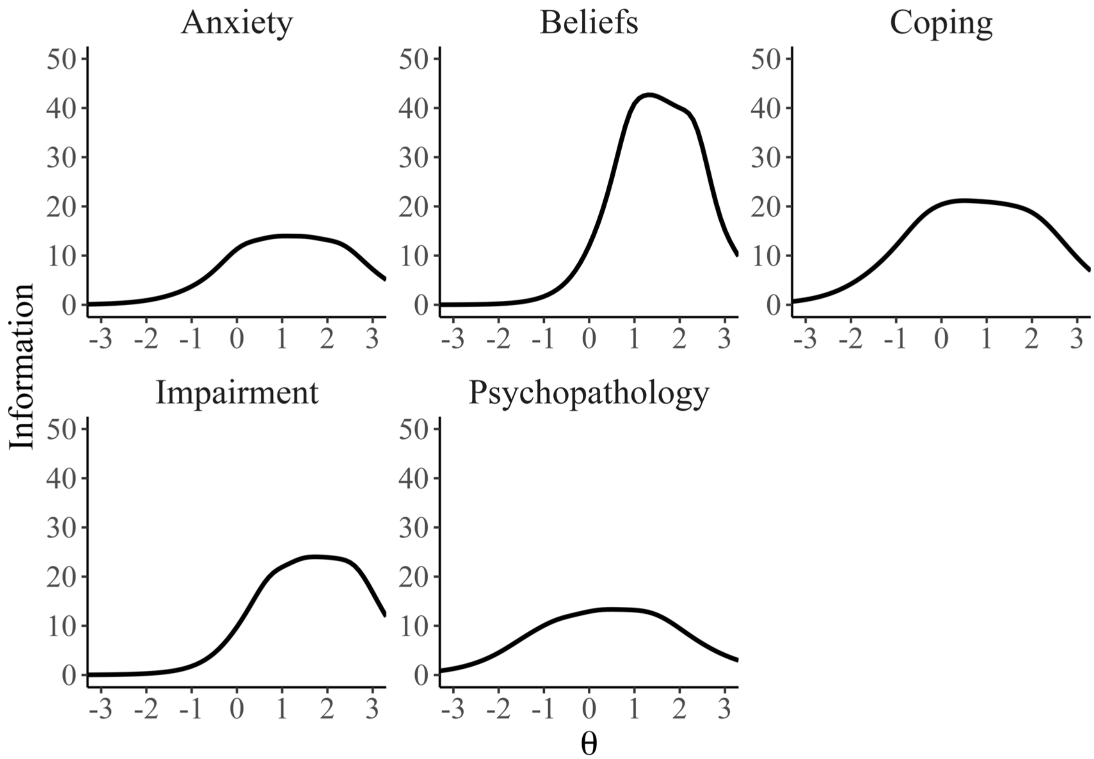

FIGURE 3 | Test information functions for the five-factor model of the 70-item DMQ using the unidimensional GRM approximations. The *x*-axis represents the latent trait (θ) values, ranging from −3 to 3, while the *y*-axis indicates the test information provided by the items within each factor. Each curve illustrates the precision of measurement across the latent trait continuum, with higher peaks reflecting greater reliability at specific levels of θ. The beliefs factor demonstrates the highest information peak around $\theta = 1.5$, indicating optimal measurement at moderate to high levels of the latent trait. In contrast, the coping factor provides maximum information near $\theta = 0$. These curves highlight the differential measurement precision of the DMQ factors across varying levels of the underlying constructs. DMQ, Duke Misophonia Questionnaire; GRM, graded response model; the individual curves represent the information function for that respective factor.

impairment, and anxiety factors were all significant predictors of the sensation avoiding factor of the ASP. As a result, the DMQ factors were not wholly divergent from general sensory processing and generalized anxiety constructs. Again, we wanted to get a sense for unexplained variance through semipartial correlations here as well. We found that 77%–80% of variance in DMQ anxiety and 78%–82% of variance in DMQ beliefs remained unexplained. Semipartial correlations for all validation measures can be found in Supporting Information S1: Table S4.

3.2 | Evaluation of the Theoretically Defined Eight-Factor Model

The authors of the DMQ constructed the scale with an eight-factor theoretical model in mind (Rosenthal et al. 2021). During development of the DMQ, this eight-factor model was never psychometrically evaluated. To provide the first psychometric evaluation of this conceptualization of the DMQ, we fit a CFA using the eight, theoretically derived factors (8F) to our full sample. The 8F model adequately fit the data, $\chi^2(2116) = 6479.990$, $p < 0.0001$; TLI = 0.926; CFI = 0.929; SRMR = 0.069; RMSEA = 0.053, 90% CI [0.051, 0.054]. The factor loadings and factor correlations for the final 8F model are reported in Supporting Information S1: Table S2. A summary of all CFA models run in this paper can be found in Supporting Information S1: Table S3. As the 8F model was derived theoretically and content analysis was performed in the original paper, we did not perform content analysis here (Rosenthal et al. 2021).

4 | Discussion

The results from our psychometric evaluation of the DMQ are twofold. First, we conducted the first complete dimensionality assessment of the DMQ. Second, we performed the first evaluation of the theoretically derived, eight-factor model of the DMQ (Rosenthal et al. 2021). As a result, we provide a novel, more parsimonious five-factor conceptualization of misophonia along with validity evidence and a detailed item-level analysis using IRT. Additionally, we provide structural evidence for the eight-factor model, supporting the theoretically derived conceptualization of misophonia from Rosenthal et al. (2021). The five-factor structure that emerged was unique to the current study and retained all 70 substantive (i.e., nontrigger) items of the DMQ. Three of the original theoretical subscales were approximately retained in terms of what facet of misophonia they measure, including the collapse of the coping subscales into one. However, the five-factor structure separated items specifically pertaining to anxiety into one factor. Misophonia is not yet formally recognized as a discrete psychiatric condition due, in part, to the overlap with other conditions, including

TABLE 3 | Regression of validation measures from the full sample onto EAP scores.

Outcome	Predictor (EAP)	Adj. R^2	*b*	*SE*	*t* value	Partial η^2
MQ						
Severity score	Beliefs	0.36	0.32	0.11	2.89*	0.27
	Coping		0.36	0.11	3.40*	0.11
	Impairment		0.43	0.11	3.78*	—
	Aversion		0.67	0.11	6.12*	0.06
	Anxiety		0.15	0.11	1.34	—
Emotions and behaviors	Beliefs	0.60	1.66	0.26	6.36*	0.51
	Coping		0.46	0.25	1.87*	0.17
	Impairment		1.42	0.26	5.40*	0.07
	Aversion		2.67	0.26	10.44*	0.16
	Anxiety		1.17	0.25	4.57*	—
Symptoms	Beliefs	0.45	0.10	0.24	0.42	0.29
	Coping		1.33	0.23	5.76*	0.21
	Impairment		0.74	0.25	2.99*	—
	Aversion		1.89	0.24	7.90*	0.10
	Anxiety		0.88	0.24	3.66*	—
ASQ	Beliefs	0.30	13.9	2.81	4.91*	0.24
	Coping		−2.12	2.65	−0.80	—
	Impairment		5.43	2.84	1.91*	—
	Aversion		7.09	2.75	2.58*	—
	Anxiety		17.08	2.77	6.18*	—
ASP						
Low registration	Beliefs	0.20	0.29	0.42	0.68	0.12
	Coping		0.48	0.40	1.19	—
	Impairment		2.88	0.43	6.73*	0.07
	Aversion		−0.8	0.42	−1.93*	—
	Anxiety		1.79	0.42	4.29*	—
Sensation seeking	Beliefs	0.01	−0.54	0.44	−1.25	—
	Coping		1.04	0.41	2.52*	—
	Impairment		0.32	0.44	0.72	—
	Aversion		−0.05	0.43	−0.11	—
	Anxiety		0.34	0.43	0.77	—
Sensory sensitivity	Beliefs	0.30	0.89	0.43	2.03*	0.21
	Coping		0.12	0.41	0.29	—
	Impairment		1.57	0.44	3.59*	—
	Aversion		1.46	0.42	3.44*	—
	Anxiety		2.48	0.43	5.82*	—
Sensation avoiding	Beliefs	0.31	1.62	0.42	3.88*	0.25
	Coping		1.25	0.39	3.18*	—
	Impairment		1.82	0.42	4.30*	—
	Aversion		0.14	0.41	0.35	—
	Anxiety		1.60	0.41	3.90*	—

Note: Effect sizes < 0.06 (medium effect; Cohen 2009) were omitted.
Abbreviation: EAP, expected a posteriori.
$^{*}p < 0.05$.

anxiety disorders (Yektatalab et al. 2022). The separation of anxiety-related items may aid research efforts working to determine whether aspects of misophonia are unrelated to generalized anxiety or represent specific experiences within the domain of generalized anxiety (Ferrer-Torres and Giménez-Llort 2022; Norris et al. 2022).

When comparing both the five- and eight-factor models, the five-factor model had a more parsimonious structure. According to Fabrigar et al. (1999), an ideal model is a simpler model that accounts for the data nearly as well as a more complex model. In addition, many psychology researchers are of the belief that good fit is necessary but not sufficient; in fact, good fit is of limited use if the model is overly complex (Browne and Cudeck 1992; Collyer 1985; Cutting 2000; Cutting et al. 1992; Pitt et al. 2002; Preacher et al. 2013; Roberts and Pashler 2000). Thus, given that the five-factor model exhibited marginally better fit and is a simpler model, we recommend its use when the primary goal is to capture the broader structure of misophonia, particularly in contexts such as screening, outcome tracking, or large-scale research.

That said, both models offer clinically meaningful insights. The eight-factor model retains more granular components, parsing responses into specific affective, cognitive, and physiological domains. This level of specificity may be advantageous for clinicians developing interventions targeting anticipatory anxiety, autonomic arousal, or cognitive misattributions, or for researchers aiming to assess how symptom components change independently over time. Moreover, scales with more narrowly defined items (e.g., coping before vs. after a trigger) can offer increased stability and sensitivity to subtle treatment effects.

In contrast, the five-factor model consolidates those components into broader factors, potentially offering a more cohesive understanding of the core of misophonia. The novel, distinct anxiety factor may have important clinical implications, as it allows for clearer differentiation between general anxiety and specific impairments associated with misophonia. Given previous findings that highlight significant overlap between misophonia and sound-related anxiety (Vitoratou et al. 2021; Yektatalab et al. 2022), this separation may support more targeted assessment of misophonia. For example, this may aid clinicians in distinguishing whether distress arises primarily from misophonia-specific impairment or from broader anxious responses to sound. This could help clarify whether symptoms reflect misophonia, co-occurring anxiety, or both, guiding more tailored treatment plans. Similarly, the aversion factor may offer clinical utility by capturing the core experiential features of misophonia, namely, the strong negative emotional and physiological responses to specific triggers. This subscale could be useful in evaluating whether interventions are effectively reducing direct aversive reactions, beyond general improvements in mood or functioning.

Ultimately, the selection of the five- versus eight-factor model should be guided by clinical or research priorities. If the goal is comprehensive symptom profiling or individualized treatment targeting, the eight-factor model may be more appropriate. If, instead, the goal is to assess global misophonia severity or track broad outcomes over time, the five-factor model offers a useful and empirically supported alternative with greater parsimony.

The current study also provided a novel perspective on the five factors through the use of MIRT which evaluated individual item information. In terms of item functioning, all 70 items provided information about the misophonia construct, and each item contributed differentially to the measurement of misophonia. Additionally, the MIRT analysis revealed that the items of the DMQ provided information focused in $\theta = [0,1]$ range. There is a considerable amount of overlap in where items provide the most information. As an overall measurement instrument, the DMQ functioned well and was informative for individuals with higher levels (i.e., greater severity) of misophonia and was least informative for those with average or below average levels of misophonia. Thus, if researchers are interested in broadening the scope of the scale, items that are informative for those with lower levels of misophonia should be developed.

The validation evidence is supportive, in part, of the five-factor model. The factors showed different patterns of association with the MQ severity scale: the anxiety and coping factors were not significant predictors. Similarly, all factors except coping were predictive of the MQ emotions and behaviors subscale. For the symptoms subscale, all but beliefs were significant predictors.

The DMQ factors were also differentially associated with broader constructs, including general anxiety (ASQ) and sensory processing sensitivity (ASP). Beliefs and anxiety were significant predictors of the ASQ, while impairment and anxiety were predictive of ASP scores. Although these associations indicate some shared variance between misophonia, anxiety, and sensory sensitivity, they also indicate partial divergent validity. Importantly, while traditional definitions of divergent validity emphasize nonsignificant associations, the modest effect sizes observed here suggest that the DMQ is capturing unique aspects of misophonia not fully related to general anxiety or sensory traits (Table 3). Estimates of unexplained variance in the anxiety factor (ranging from 77% to 80%) and beliefs factor (ranging from 78% to 82%) also support that these constructs are unique.

This distinction becomes clearer when comparing the content of the DMQ items with those from the ASQ and ASP. For example, while all three measures include items related to anxious responses, the DMQ specifically targets distress in response to sound triggers. Items like, "When intensely bothered by a sound or sounds, please rate how often [you] became rigid or stiff" captures misophonia-specific reactions. In contrast, the ASQ includes more generalized items like "How frequently have you experienced worrying in the past week" which does not reference any sensory or contextual cues. A similar distinction can be made between the DMQ and the ASP. While an item such as "I startle easily from unexpected or loud noises" reflects general sensory sensitivity, DMQ items assess emotional reactions to specific sounds (e.g., "I am helpless"), which are central to misophonia. These item-level differences suggest that, although related, misophonia involves a unique pattern of responses that do not fully explain existing anxiety or sensory sensitivity frameworks.

Together, these findings support the idea that misophonia is conceptually distinct, even if partially overlapping, with related constructs. Future research using methods like multitrait-multimethod designs or longitudinal studies could assist in clarifying these boundaries. The current results provide initial evidence that the DMQ measures clinically meaningful features of misophonia that are not redundant with anxiety or sensory processing difficulties.

The current study aimed to establish the factor structure of the DMQ through a combination of psychometrics and theory, as no previous study examined the factor structure of the DMQ through use of both EFA and CFA. Given that this is the first study to test and empirically support a five-factor model of the DMQ, subsequent replications of this structure would add to the confidence of the five-factor model identified in the present study. Although the five-factor model shares similarities with the theoretically derived factors in the original scale development paper (e.g., the beliefs factor remains the same), additional work should be done to examine the five-factor structure in a clinical sample where all or the majority of respondents have clinically determined misophonia (Rosenthal et al. 2021).

Future work would also benefit from an evaluation of differential item functioning (DIF). Given that clinical disorders often present differently across groups (e.g., males vs. females), there is reason to hypothesize that DMQ items would function differently across groups (Salk et al. 2017; Schlechter et al. 2024). DIF analyses could help determine whether individuals with similar levels of misophonia respond differently to specific items due to group membership (e.g., gender), rather than actual differences in symptom severity. This is particularly important when interpreting subscale scores, such as anxiety or impairment, where demographic-linked interpretation biases could skew conclusions about severity or treatment needs. Future research should consider both item-level DIF testing and structural models (e.g., MIMIC) to formally assess measurement invariance. A MIMIC framework would allow for adjustments of one exogenous variable (e.g., gender) while accounting for the effects of another (e.g., age), offering a more nuanced understanding of whether observed group differences reflect true variance in misophonia or measurement bias. Establishing invariance would be a critical step toward ensuring that the DMQ performs accurately across diverse populations.

In addition, future work may benefit from using exploratory structural equation modeling (ESEM) to evaluate the eight- and five-factor models. Unlike typical applications of structural equation modeling where cross-loadings are constrained to zero, ESEM allows for cross-loadings to be freely estimated. This is particularly relevant in this context, where several items appeared to shift meaningfully across factors in the transition from the theoretical DMQ structure. This approach could offer further insights into the multidimensional structure of misophonia and clarify whether certain items systematically load onto multiple latent traits, potentially reflecting the interconnected nature of emotional, physiological, and cognitive responses to sound.

In terms of limitations, this study only uses self-report measures. Future studies of the DMQ would benefit from including measures beyond self-report. For example, validation of the DMQ using biological techniques to evaluate physiological differences would provide additional support for continued use of the scale. Specifically, validating the DMQ using measures with known differences in neurophysiology and physiological responses to trigger exposure would add strength to the measure and any future modifications to the DMQ (Cerliani and Rouw 2020; Kumar et al. 2021). The field would also benefit from further interdisciplinary work on misophonia, as the combination of self-report measures with biological measures can achieve a more comprehensive view and understanding of the misophonia construct.

While the use of a nonclinical, university-based sample limits the clinical utility of our findings, particularly in the absence of formal misophonia diagnostic criteria, this approach reflects necessary trade-offs in early-stage scale development. Obtaining a clinical sample would require administering clinical interviews (e.g., Duke Misophonia Interview; Guetta et al. 2022), which for a sample of the current size would require significant clinical resources and consortium-style recruitment. As such, our sampling strategy aligns with prior work involving early-stage scale development (e.g., Baker et al. 2019; Rosenthal et al. 2021; Vitoratou et al. 2021), where large convenience samples are often prioritized to ensure adequate power for factor analyses and model replication.

Misophonia is likely best understood as a spectrum of experiences, some of which fall within the typical (i.e., nonpathological) range of emotional responses. Clarifying where on the spectrum these emotional responses and misophonia-related experiences become clinically significant and indicate the need for clinical care requires research that captures a wide range of symptom severity, including responses that fall below the threshold for clinical care. The present study contributes to this effort by examining patterns that may help define the threshold for clinical relevance. It also highlights the ongoing challenge of distinguishing misophonia from overlapping conditions like anxiety, an issue reflected in our findings. Identifying the core features of misophonia that drive meaningful disruption in daily life is essential for guiding future diagnostic decisions. While our sample may underrepresent individuals with highly severe symptoms, large-scale studies like ours are needed to move toward identification of these features. Future work should incorporate both clinical and general-population samples to continue validating the DMQ and to assess whether it functions differently in samples from a general population than it does in samples from a clinical population.

As for the validity measures used in this study, only one measure of misophonia was used (the MQ) to assess convergent validity. Although this measure was chosen as it is one of the most commonly used measures in the literature, the alignment of the DMQ factors with other measures such as the Amsterdam Misophonia Scale or the Misophonia Assessment Questionnaire should be examined in future evaluations of the DMQ five-factor model (Johnson 2014; Schröder et al. 2013). Although these measures were included in the initial validation of the DMQ, further work examining the five-factor structure with additional measures of misophonia could be an important contribution to the DMQ literature (Rosenthal et al. 2021)

To conclude, this study provides the first complete psychometric investigation, to our knowledge, of the *overall* structure of the DMQ substantive items. As the DMQ is the first psychometrically validated self-report measure of misophonia, we believe the results we present in support of its empirically and theoretically derived factor structures only strengthens the evidence in support for the DMQ's use in misophonia research.

Author Contributions

Catherine M. Bain conceptualized the project, selected methodology, oversaw and conducted the project (i.e., data collection and curation), conducted the analyses (i.e., formal and coded), and generated all figures. Jordan E. Norris oversaw and conducted the project (i.e., data collection and curation), contributed to methodology, contributed to the analyses (formal analysis only), and acquired/developed project resources. Both Catherine M. Bain and Jordan E. Norris drafted the initial manuscript draft. Alex Conley helped with both data collection and curation. Analyses were validated by Patrick D. Manapat. Patrick D. Manapat and Lauren E. Ethridge both supervised the project. All authors contributed to the writing, review, and editing of the manuscript draft and approve of the final manuscript. The project did not require funding.

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The authors have nothing to report.

Ethics Statement

The Institutional Review Board at the University of Oklahoma, Norman approved our interviews (IRB: 17681). Respondents gave online informed consent before starting the survey.

Consent

As a nonclinical sample was used, patient consent is not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data are available upon reasonable request to the corresponding author.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.
SupplementaryMaterials.